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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.C.A. (CBCS) Degree Examinations

Supplementary Examination - December 2025

22MCA103 – COMPUTER ORGANIZATION AND ARCHITECTURE

Duration: 3 Hours

Max. Marks:100

Note: Answer **One full** question from each unit.

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are basic gates? Realize the basic gates using NAND gate and NOR gate.	12	L*3	1	2
	b) Simplify $Z = AB + (AC)' + A B'C(AB+C)$ using Boolean laws.	08	L3	1	1
OR					
2.	a) Realize XOR and XNOR gates using NAND gates.	10	L3	1	2
	b) Convert the following:				
	i) $(1792)_{10}$ in Binary				
	ii) $(110101)_2$ into decimal.				
	iii) $(E7)_{10}$ into Binary				
	iv) $(152)_8$ into Binary				
	v) $(275)_8$ into Hexadecimal.	05	L3	1	1
	c) State and explain De-morgan Laws.	05	L3	1	2
Unit – II					
3.	a) Simplify the following expression using K maps. Also show the intermediate steps:				
	i) $F(A,B,C,D) = A'BC'D' + A'B'C'D + A'B'CD + A'BCD' + AB'C'D + AB'CD$				
	ii) Minimize the following function in POS form				
	$F(A,B,C,D) = m(0,1,2,6,7,8,13,14,15) + d(3,5,12)$	10	L3	2	3
	b) What is the use of the Magnitude comparator? Design a 2 bit Magnitude comparator, highlighting the truth table and further simplification.	10	L2	2	2
OR					
4.	a) Simplify the following expression using Tabulation method:				
	$F(A,B,C,D) = m(6,7,8,9) + d(10,11,12,13,14)$	12	L3	2	3
	b) Implement the following Boolean function using 8 X 1 Multiplexer,				
	$F(A,B,C,D) = m(1,3,4,11,12,13,14,15)$	08	L2	2	2
Unit – III					
5.	a) Perform non -restoring division on the numbers $15 / 4$.	10	L3	3	2
	b) Perform the following operations on 5 bit signed numbers using 2's complement representation system. State whether overflow occurs or not in each case.				
	i) $(-10) + (-13)$ ii) $(-10) - (+4)$ iii) $(-3) + (-8)$				
	iv) $(-10) - (+7)$ v) $(+4) + (-6)$	10	L2	3	2
OR					
6.	a) Given Booth's algorithm to multiply two binary numbers, explain the working of the algorithm by multiplying the multiplicand +21 with multiplier -18.	10	L3	3	2
	b) Explain the design of a 16-bit carry look ahead adder built from four 4-bit adders.	10	L1	3	1

P.T.O

Unit - IV

- a) What is addressing mode? Explain any five addressing modes with appropriate examples.
- b) Explain Big Endian and Little Endian assignment. Consider a computer that has a byte addressable memory organized in 32 bit words according to the Big Endian scheme. The memory starts at address 3000. Show the contents of three memory words at locations 3000, 3004 and 3008 after the string "NITTE UNIVERSITY" is entered.
(ASCII Codes: N- 78, I - 73, T- 84, E-69, U-85, V-86, R-82, S-83, T-84, Y-89, Space =32)

12 L2 4 1

OR

08 L2 4 2

8. a) Draw and explain the connection between memory and processor via the respective registers.
- b) Define Subroutine and parameter passing. Explain how to pass parameters by value and by reference with examples.

12 L2 4 1

08 L2 4 2

Unit - V

9. a) What is cache? Illustrate the different mapping functions in cache.
- b) Explain the methods of handling interrupt requests from multiple I/O devices.
- OR
10. a) Discuss the organization of a 1 K x 1 memory chip.
- b) With a timing diagram, explain the working of synchronous bus.

10 L1 5 1

10 L1 5 1

10 L1 5 1

10 L1 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
